

MUHAMMAD AMMAR

Software Engineer — Machine Learning Enthusiast — Entrepreneur

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EDUCATION

Bachelor of Science in Software Engineering — GIKI 2023 – Present
A-Levels — Jinnah Highs Grades: 2 B's **O-Levels (Cambridge IGCSE)** — Jinnah Highs Grades: 2 A*'s and 5 A's

EXPERIENCE

Spectra Ops – Co-Founder & Country Head 2024 – Present

- Co-founded a technology solutions company; leading Pakistan operations, client relationships, and business development
- Overseeing technical strategy, team coordination, and building partnerships with local and international clients

Safar-e-GIKI – Founder & Operations Manager 2024 – Present

- Founded and managing a transport booking service for GIKI students traveling to major cities (100+ bookings/month)
- Built and deployed the booking platform; managing seat inventory, payment tracking, trip scheduling, and vendor coordination

PROJECTS

Mission Dossier: MIT-05 — Agentic Healthcare Intelligence System [GitHub](#)

- Built at the 5th Global MIT Hackathon; forked from original team repo and extended with additional agents and UX refinements
- Designed an agentic AI system pairing a patient-facing intelligence layer with a clinician copilot for healthcare workflows
- Autonomous agents coordinate clinical reasoning across symptom intake, triage, and treatment recommendations
- Implemented conversational interface powered by LLMs with grounded retrieval and tool use; modular, composable architecture
- Stack: React, TypeScript, Agentic AI, LLM Tool Use

Wheels Predict – Car Price Prediction System [GitHub](#)

- Built end-to-end ML pipeline for Pakistani used car price prediction using 50,000+ listings scraped from PakWheels via Scrapy
- Implemented XGBoost and Random Forest models (85%+ accuracy); Flask web interface with real-time predictions and monitoring
- Containerized with Docker; deployed CI/CD pipeline using GitHub Actions

Safar-e-GIKI – Bus Ticket Booking Platform [Live Demo](#)

- Full-stack bus booking platform for GIKI students with seat selection, real-time booking, payment integration, and admin analytics
- Built with React, TypeScript, and Supabase for real-time database and authentication

HCI/UI-UX Design Project

- Full UX process: user research, sketches, wireframes, high-fidelity Figma prototypes, and UML diagrams (use case, class, sequence)

Virtual Qari Memory System [GitHub](#)

- Quran memorization app with progress tracking, personalized learning paths, and audio playback features

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, HTML, CSS, SQL

Frontend: React, Vite, Tailwind CSS, Radix UI, Recharts, Responsive Design

Backend & Databases: Supabase, PostgreSQL, Flask, FastAPI, RESTful APIs

ML & Data: PyTorch, Scikit-learn, Pandas, NumPy, XGBoost, Agentic AI, LLM Tool Use, Data Visualization

Web Scraping & Automation: Scrapy, BeautifulSoup, Selenium, Data Pipelines

DevOps & Tools: Git, GitHub, Docker, GitHub Actions, CI/CD, Vercel

Design & UX: Figma, UML Diagrams, Wireframing, Prototyping, User Research

CERTIFICATIONS

SQL (Udemy)